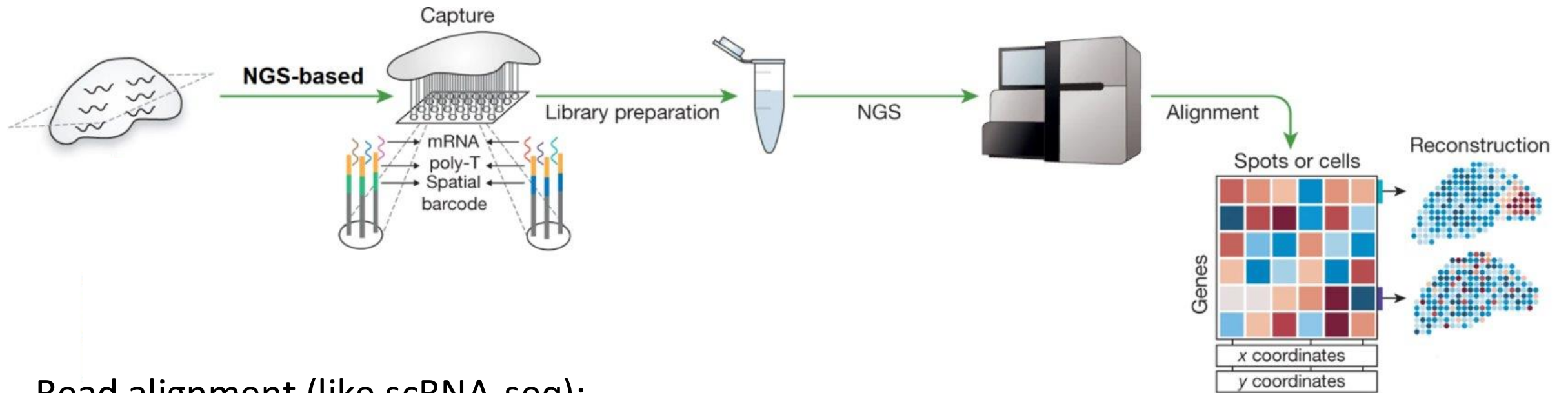


Preprocessing sequencing-based spatially-resolved transcriptomic (SRT) data

Ahmed Mahfouz

Human Genetics, Leiden University Medical Center
Pattern Recognition and Bioinformatics, TU Delft

Sequencing-based SRT

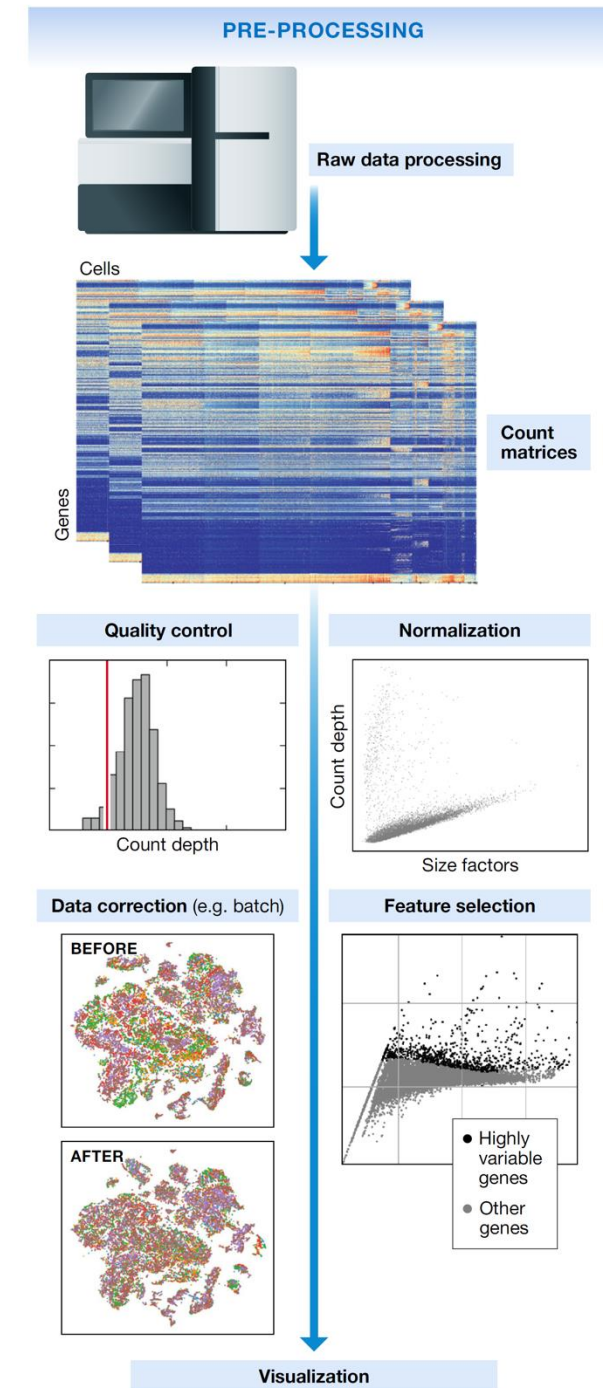


Read alignment (like scRNA-seq):

- SpaceRanger from 10x Genomic
- Stereo-seq Analysis Workflow SAW from STOmics
- ...

Preprocessing SRT data

- Like scRNA-seq data preprocessing:
 - Reads to count matrix
 - Quality control (QC)
 - Normalization
 - Batch correction
 - Feature selection



Factors that affect data quality in scRNA-seq

Cell dissociation

Cell capture

Cell lysis

Reverse transcription

Preamplification

Library preparation and sequencing

Factors that affect data quality in seq-based SRT

Cell dissociation

Cell capture

Cell lysis/permeabilization

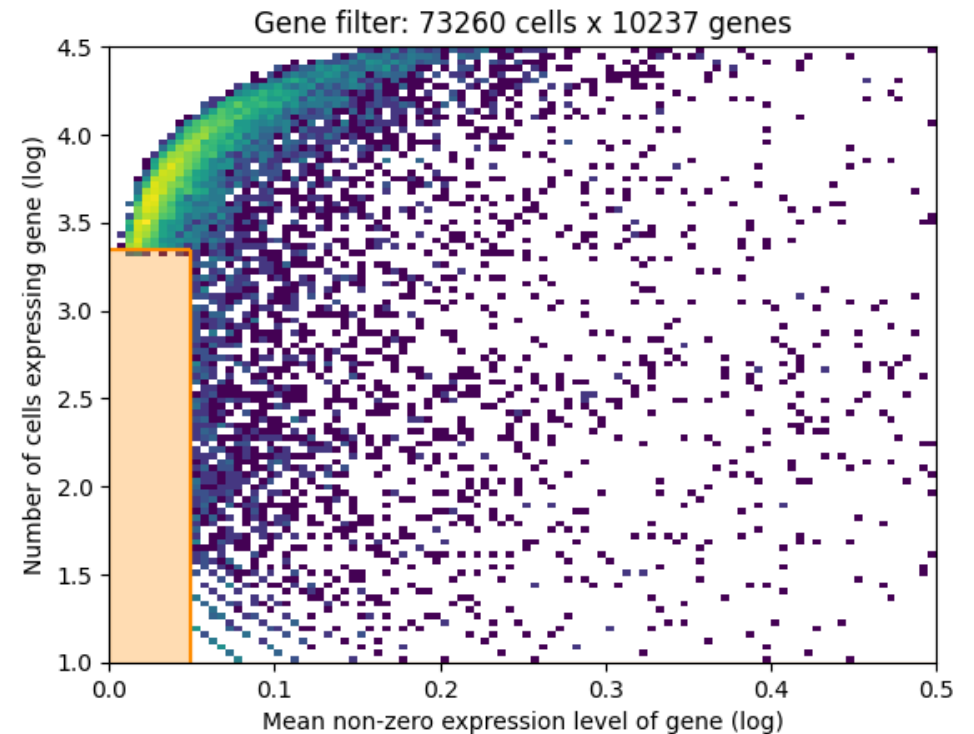
Reverse transcription

Preamplification

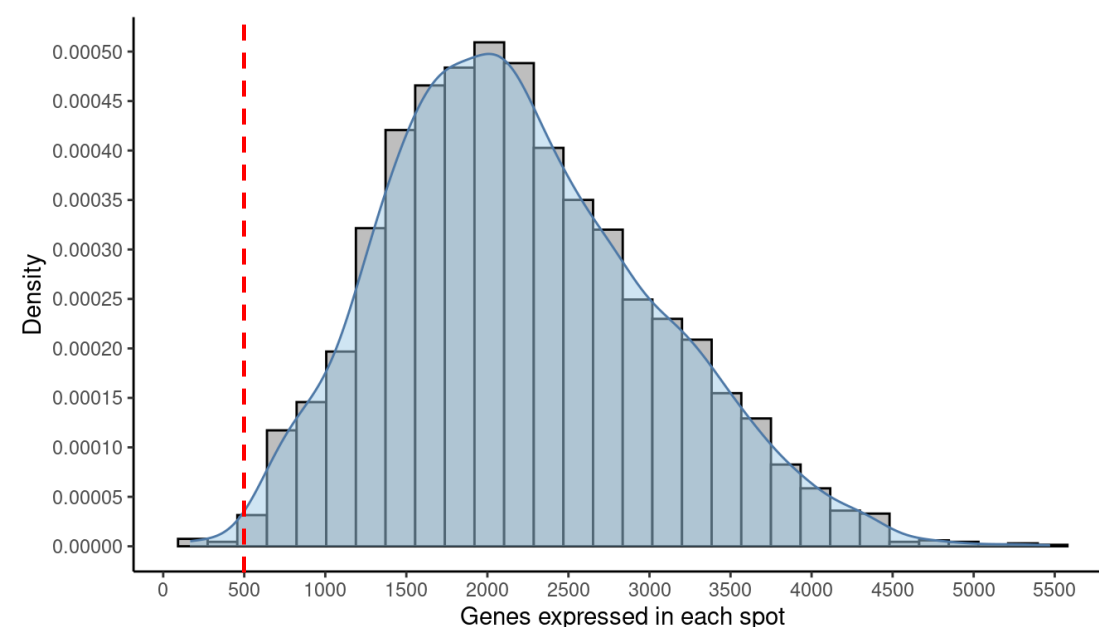
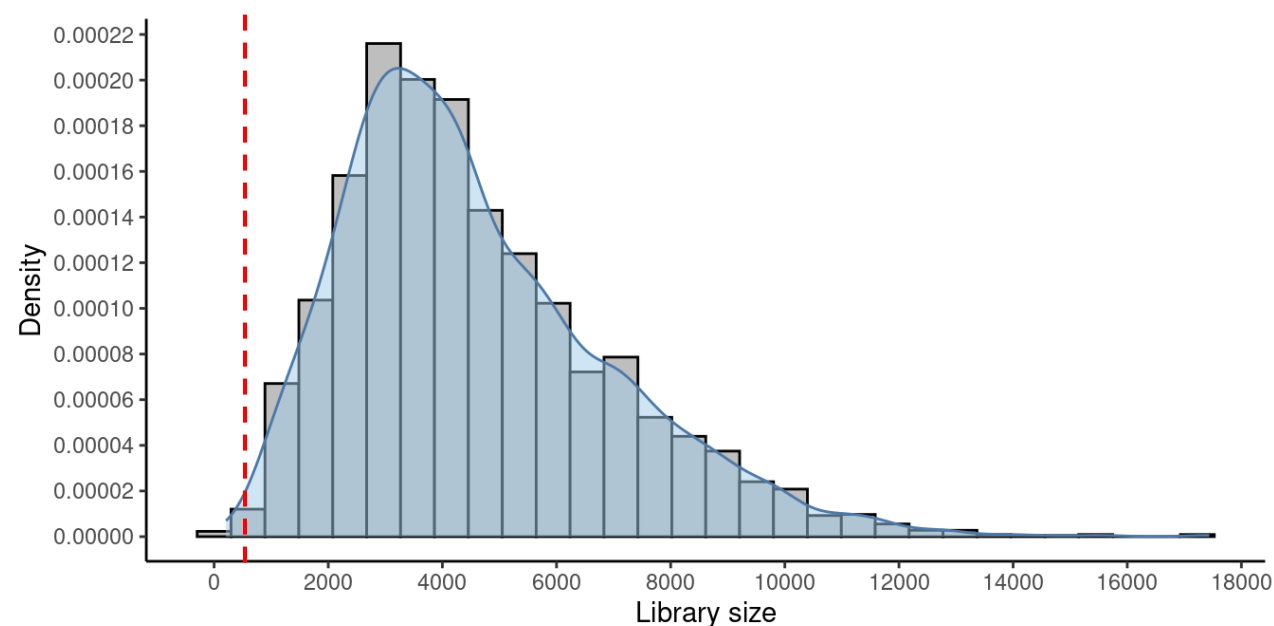
Library preparation and sequencing

Filtering genes

- Remove mitochondrial genes (gene names start with prefix mt- or MT-):
 - They compose 15-40% of mRNA in each location
 - Mostly representing technical artifacts
- Filter based on cell count, cell percentage, non-zero mean (aim at 8-16 k genes)

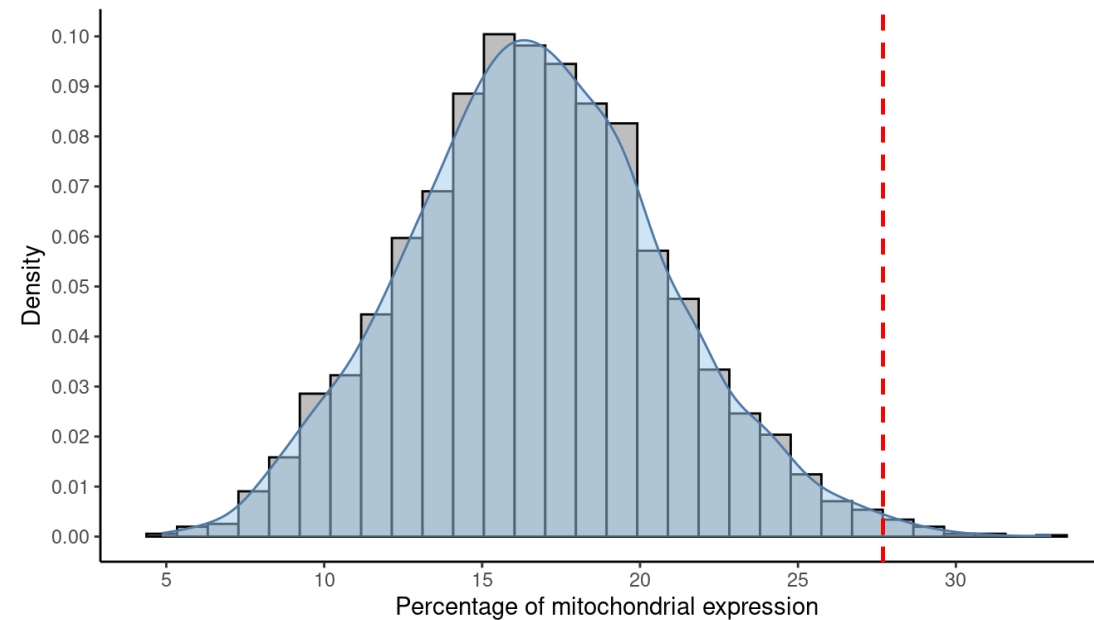


Filtering spots (counts/genes)



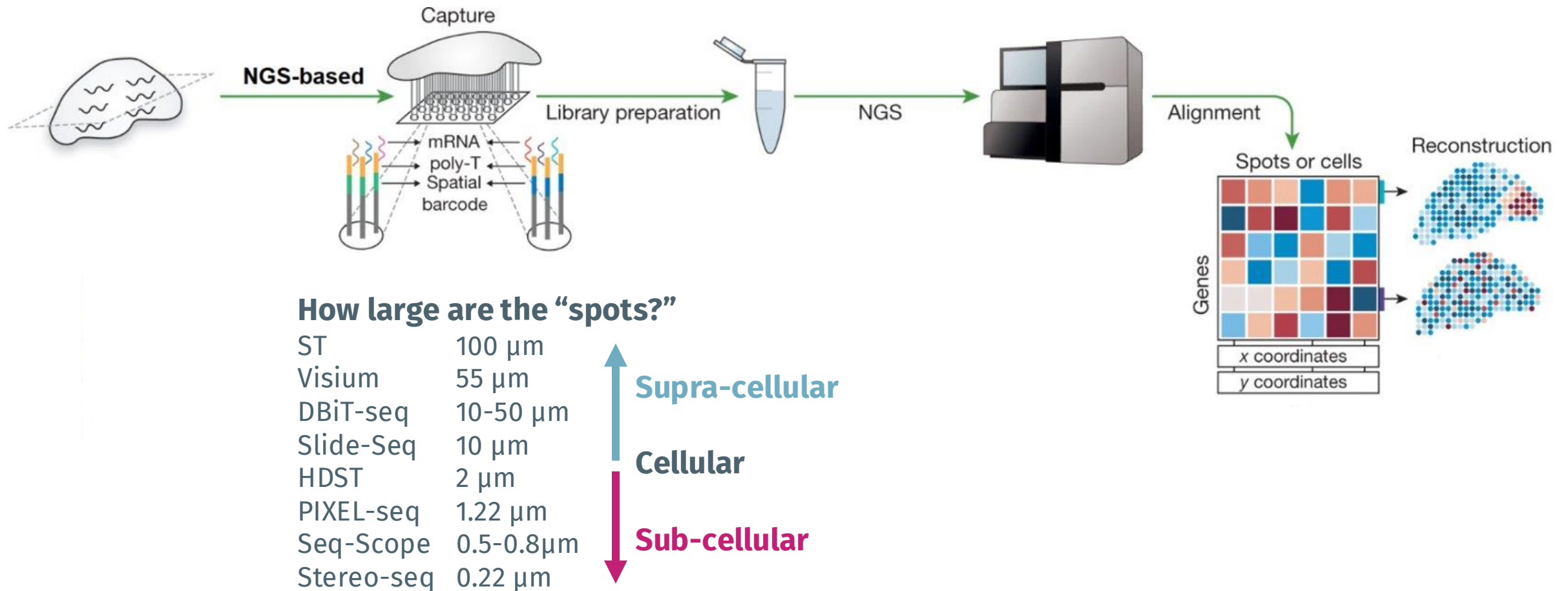
Poor mRNA capture rates due to cell damage and missing mRNAs, or low reaction efficiency

Filtering spots (mitochondrial %)



Partial cell lysis leading to leakage and missing cytoplasmic mRNAs, with the resulting reads therefore concentrated on the remaining mitochondrial mRNAs

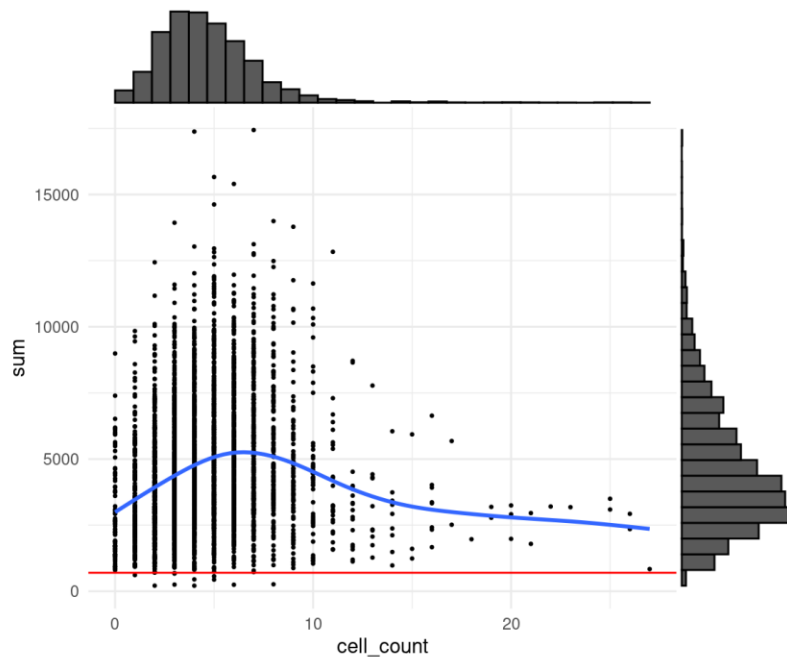
Resolution variation in sequencing-based SRT



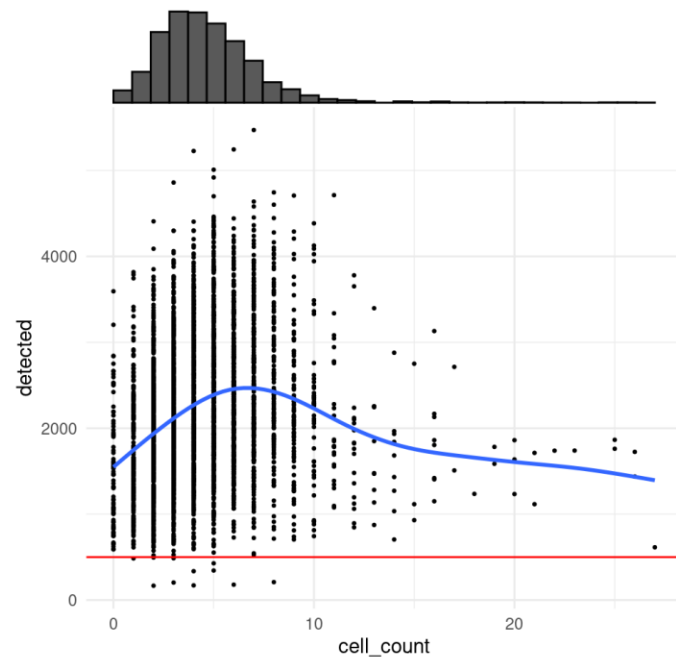
Spots may contain zero, one, or multiple cells

We can plot the library sizes against the number of cells per spot (to make sure we don't remove any spots that may have biological meaning)

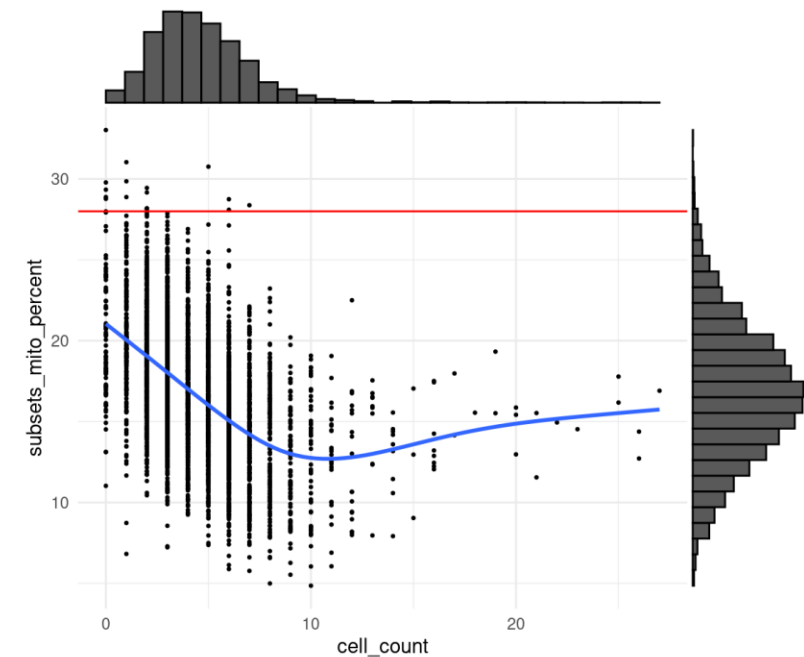
Library size



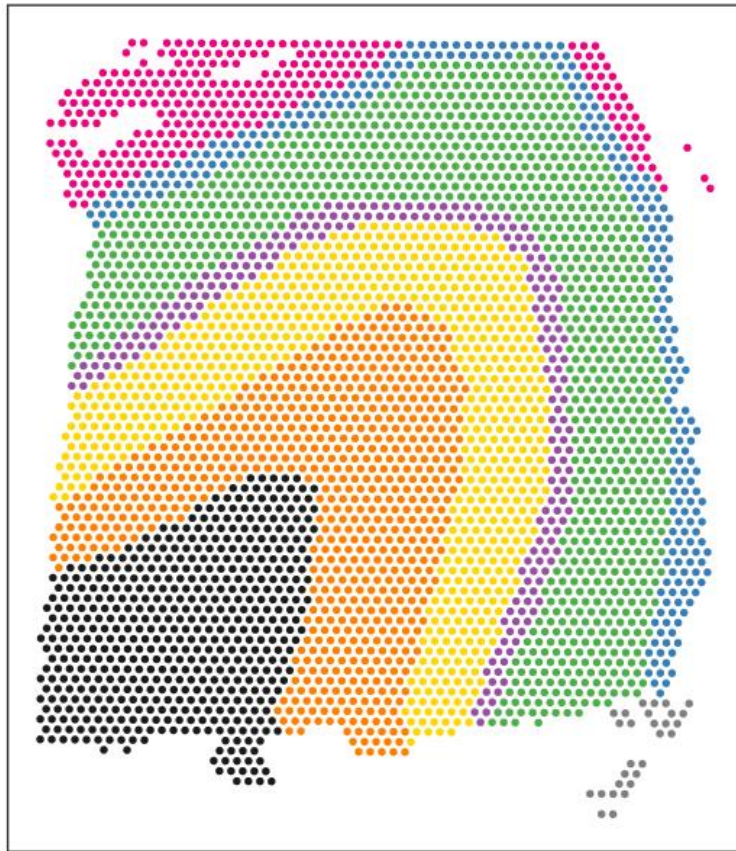
Total Genes



Mitochondrial %

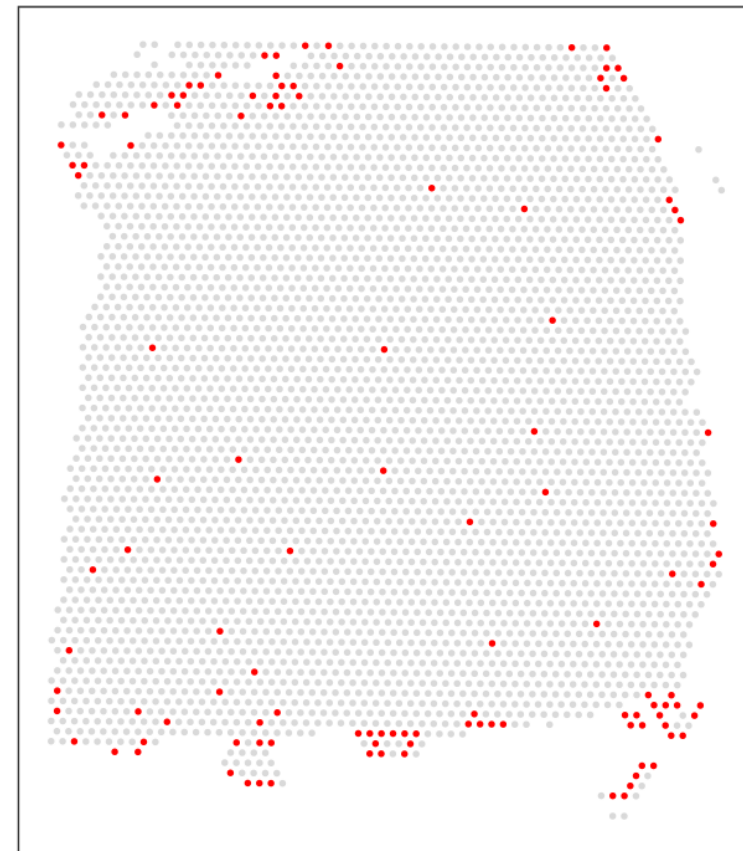


Check spatial patterns



ground_truth

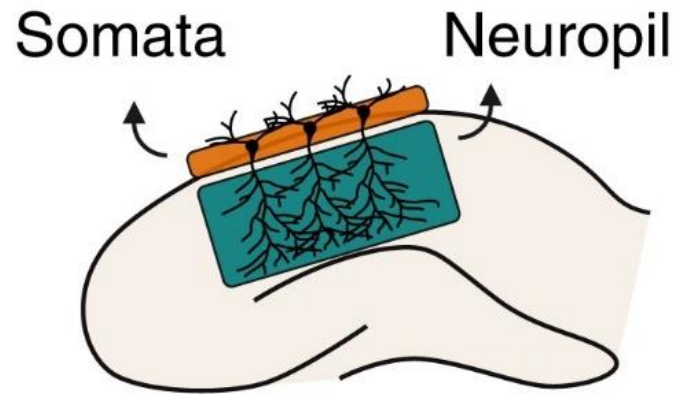
- Layer1
- Layer2
- Layer3
- Layer4
- Layer5
- Layer6
- WM
- NA



discard

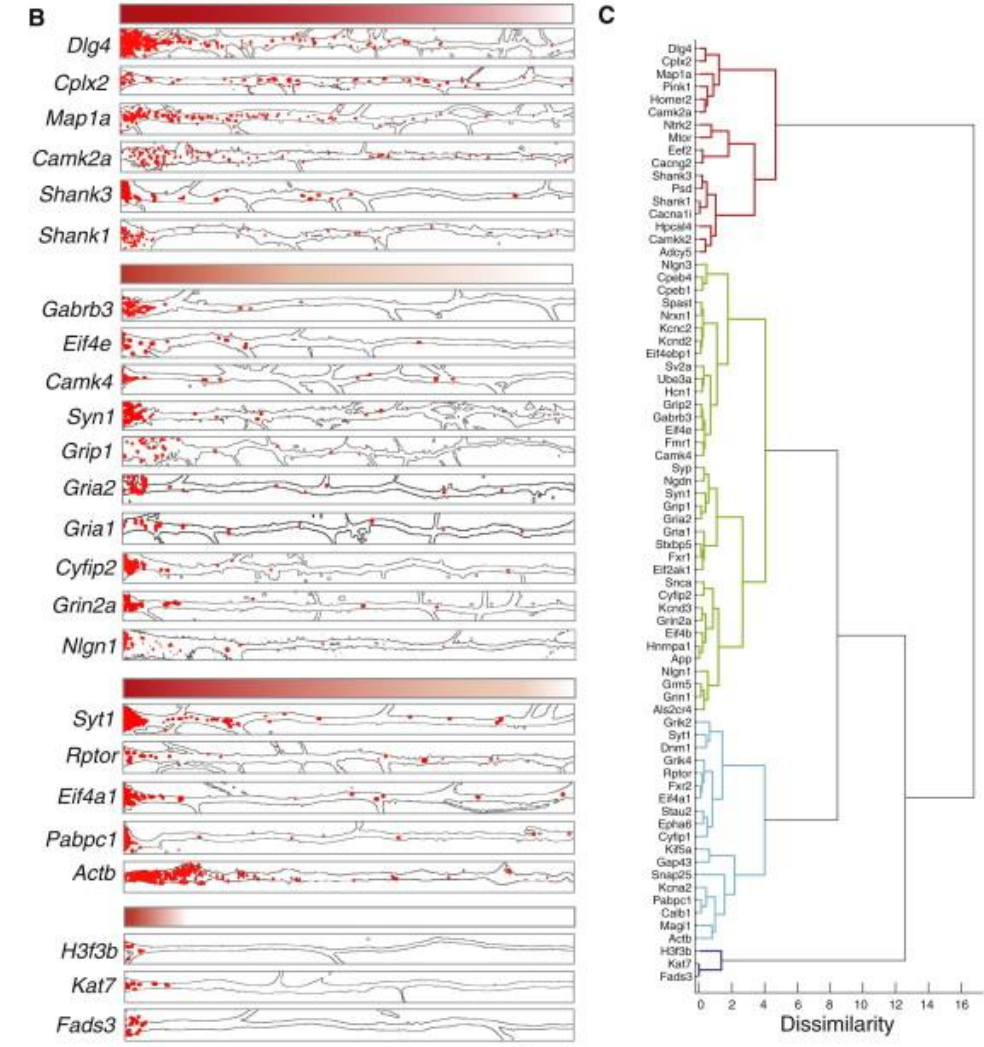
- FALSE
- TRUE

mRNA localization within cells



SUMMARY

In neurons, dendritic protein synthesis is required for many forms of long-term synaptic plasticity. The population of mRNAs that are localized to dendrites, however, remains sparsely identified. Here, we use deep sequencing to identify the mRNAs resident in the synaptic neuropil in the hippocampus. Analysis of a neuropil data set yielded a list of 8,379 transcripts of which 2,550 are localized in dendrites and/or axons. Using a fluorescent barcode strategy to label individual mRNAs, we show that their relative abundance in the neuropil varies over 3 orders of magnitude. High-resolution in situ hybridization validated the presence of mRNAs in both cultured neurons and hippocampal slices. Among the many mRNAs identified, we observed a large fraction of known synaptic proteins including signaling molecules, scaffolds and receptors. These results reveal a previously unappreciated enormous potential for the local protein synthesis machinery to supply, maintain and modify the dendritic and synaptic proteome.



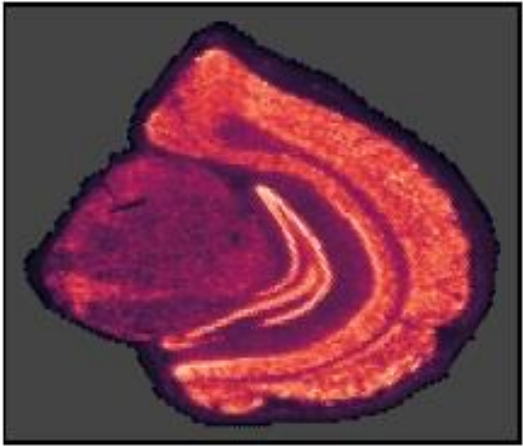
QC in sc/snRNA seq vs SRT

- sc/snRNA-seq: mRNA transcripts from a cell body or nucleus
- SRT: mRNA transcripts from a wide variety of biological domains
- SRT: tissue architecture can result in differences in reagent permeability
 - This would lead to sampling differences across the tissue and subsequently library size differences

Detection density and total detections/library sizes are associated with biology

STOmics (Mm Brain)

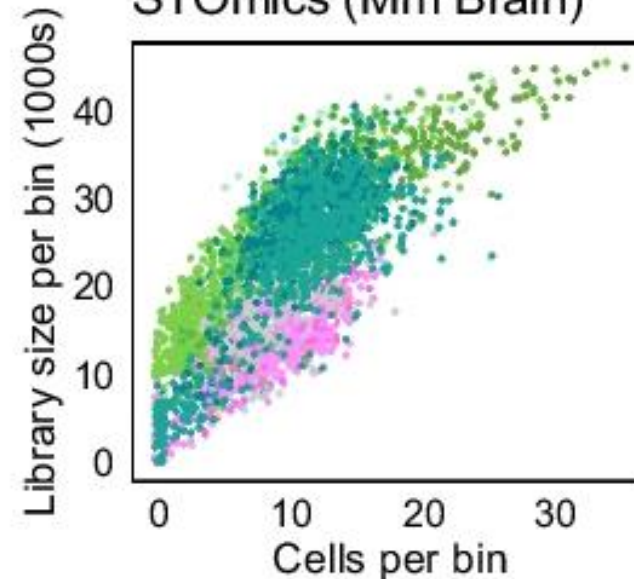
Library size per bin: 0–26.5k



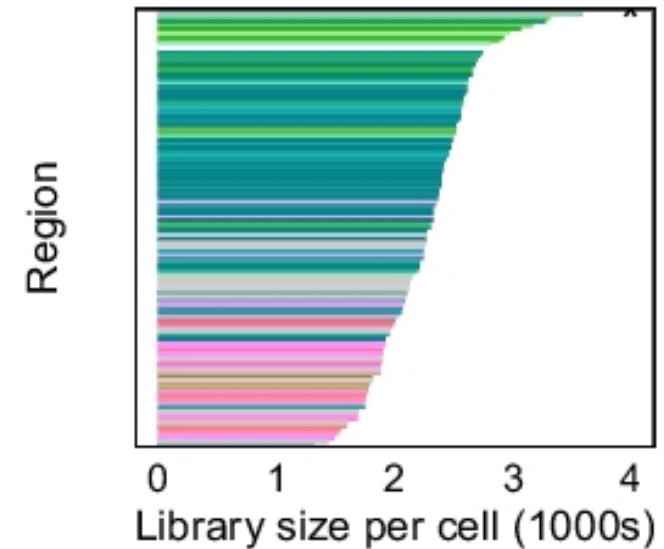
STOmics (Mm Brain)



STOmics (Mm Brain)



STOmics (Mm Brain)

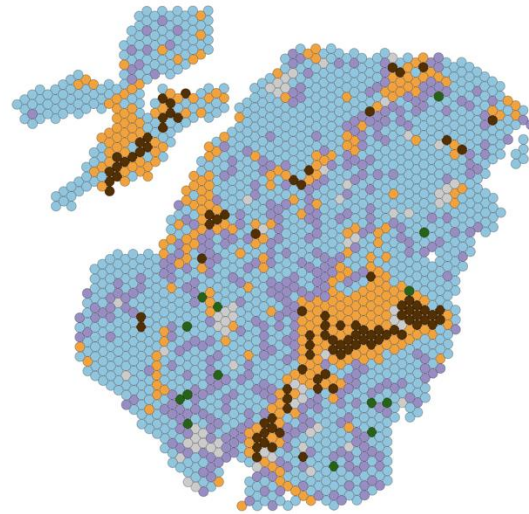


Be aware, variation in QC measure can be biological

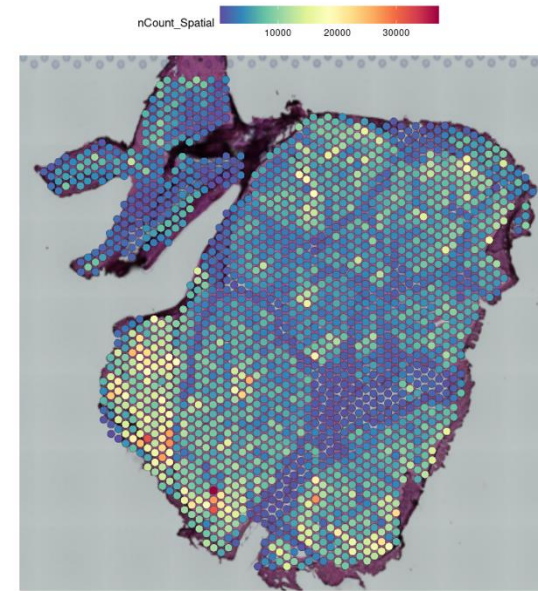
H&E



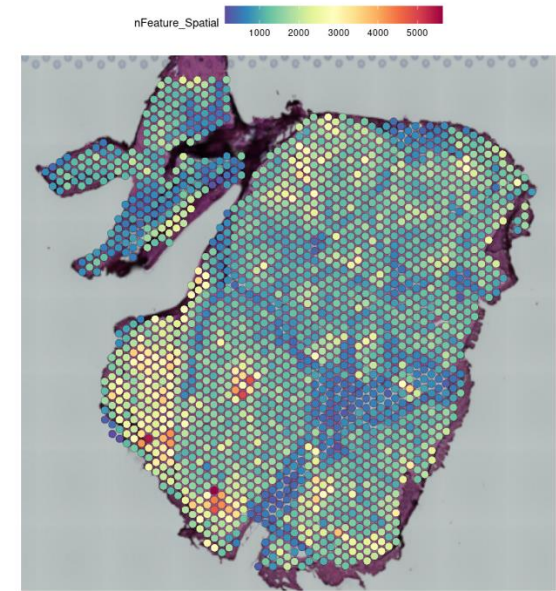
Annotation



Library size

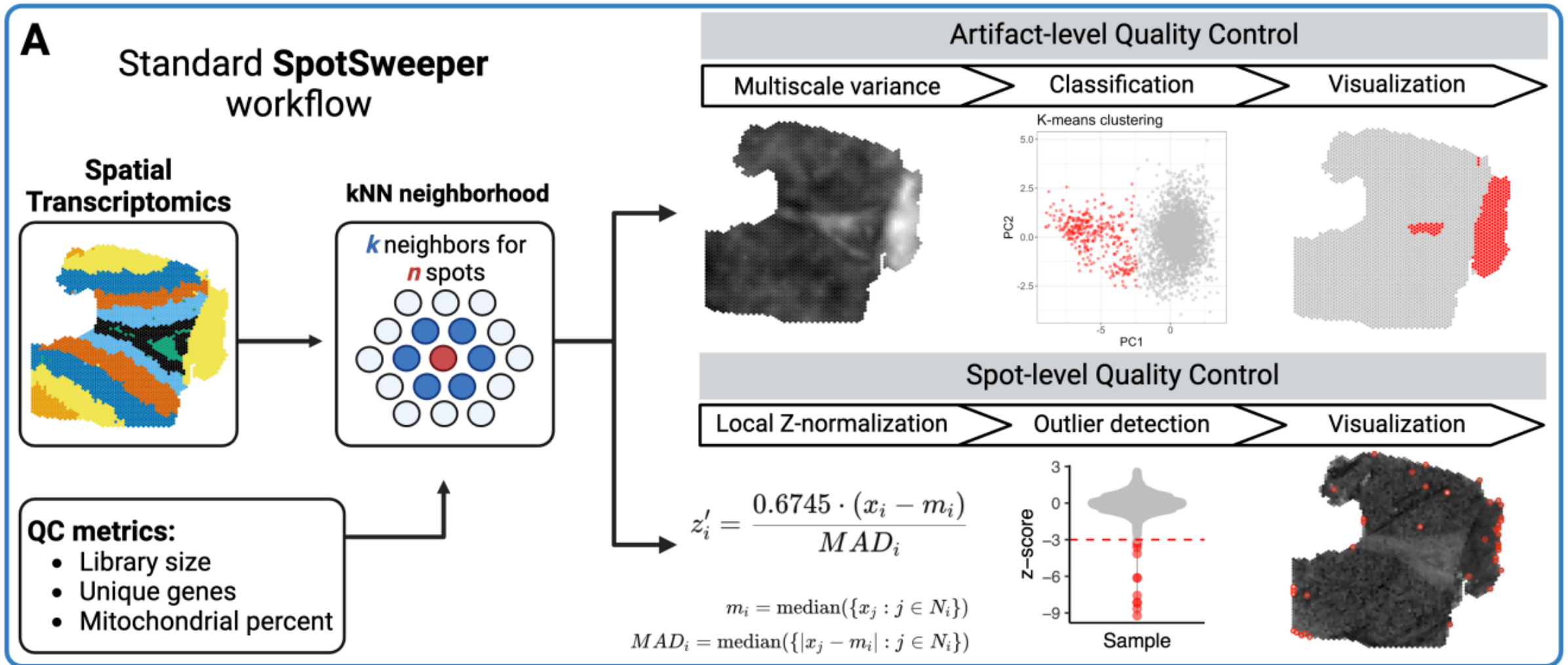


Total Genes



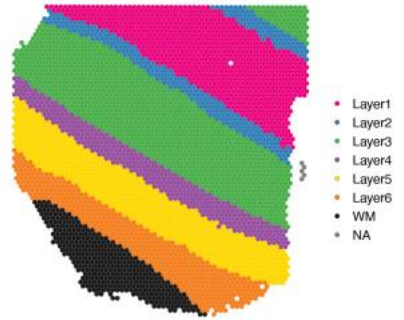
- Type 1
- Type 2A
- Type 2X
- Connective tissue
- Fat
- Non muscle fibers

Spatially-aware QC (SpotSweeper)

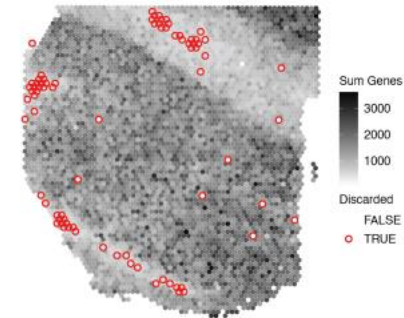


SpotSweeper (spot-level artifacts)

Annotation

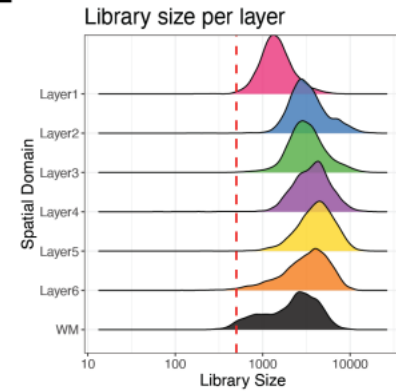


Global

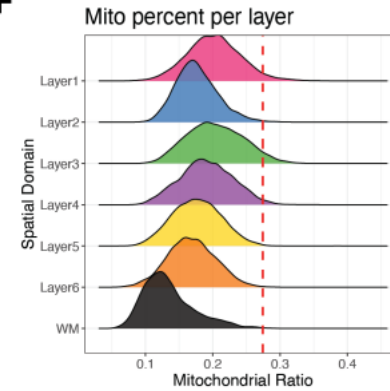


Global outliers

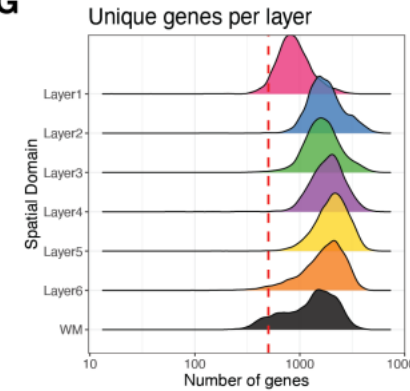
E



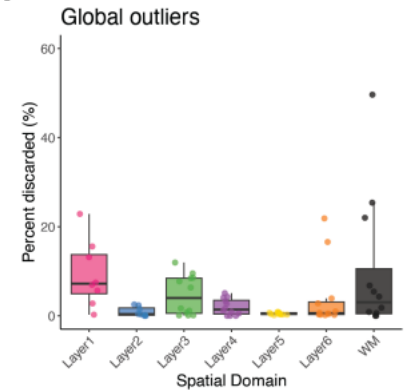
F



G



H

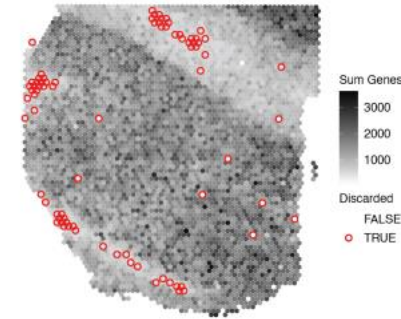


SpotSweeper (spot-level artifacts)

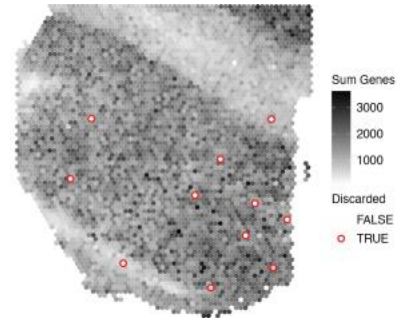
Annotation



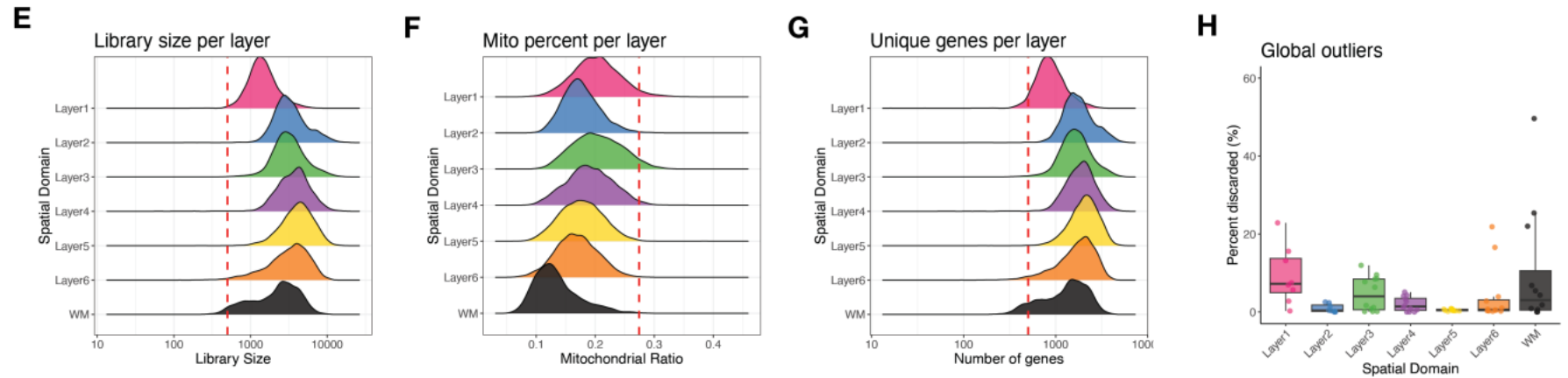
Global



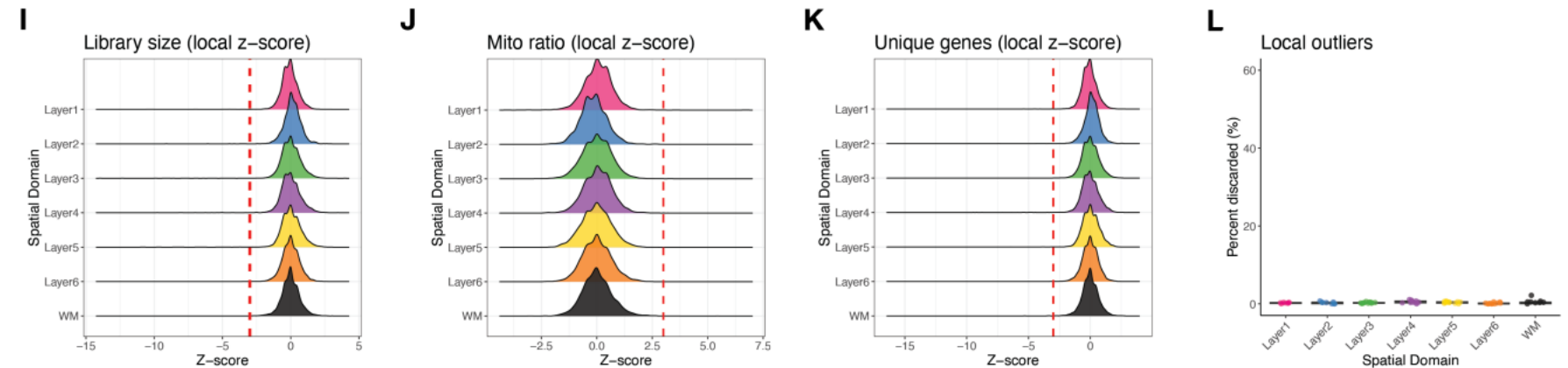
Local



Global outliers

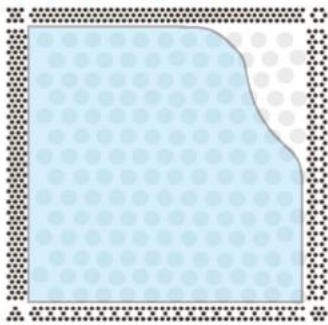


Local outliers



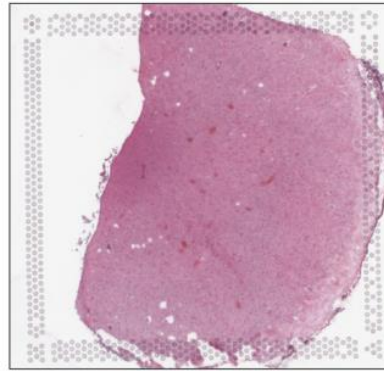
SpotSweeper (region-level artifacts)

Incomplete coverage of Visium array
Liquid reagent

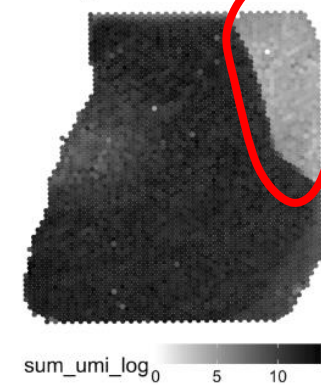


Dry spots result in
smaller library size
and fewer genes
detected

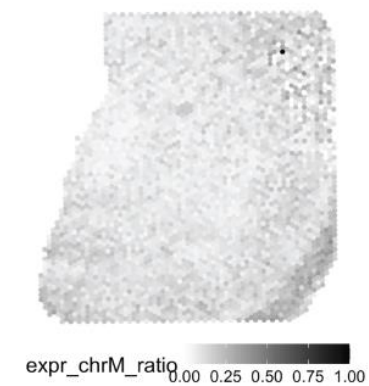
Br3942_mid



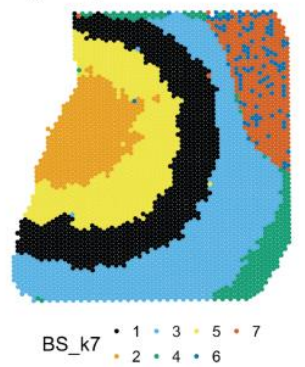
Library size



Mito Ratio



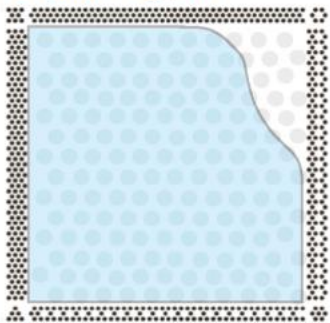
BayesSpace k=7



SpotSweeper (region-level artifacts)

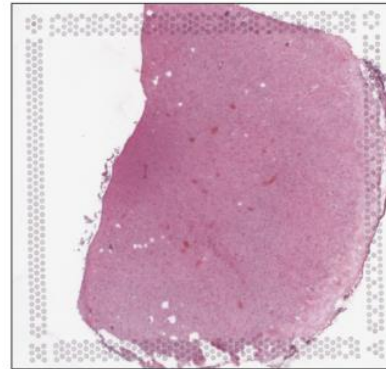
Incomplete coverage of Visium array

Liquid reagent

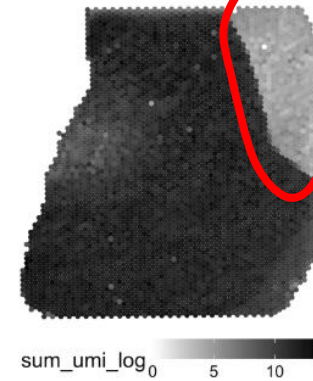


Dry spots result in smaller library size and fewer genes detected

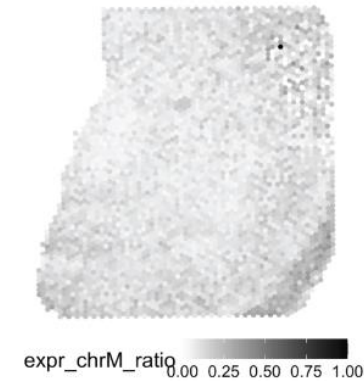
Br3942_mid



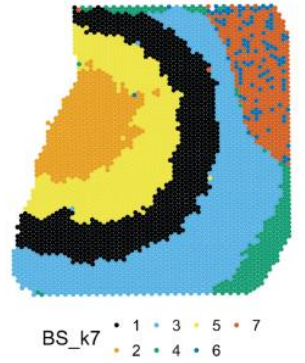
Library size



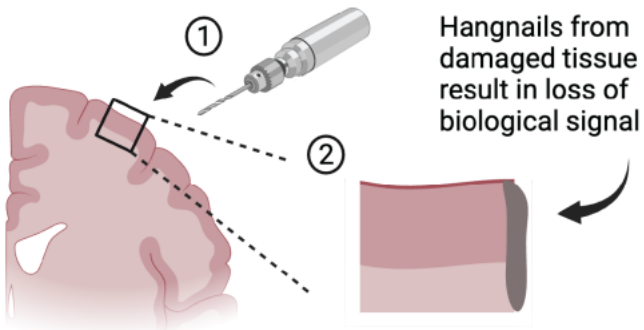
Mito Ratio



BayesSpace k=7



Tissue damage during dissection

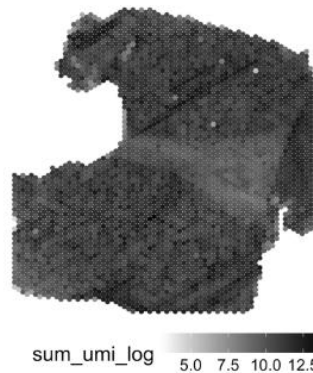


Hangnails from damaged tissue result in loss of biological signal

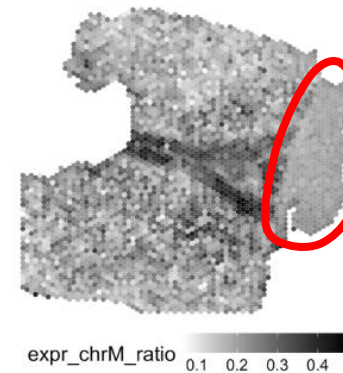
Br8325_ant



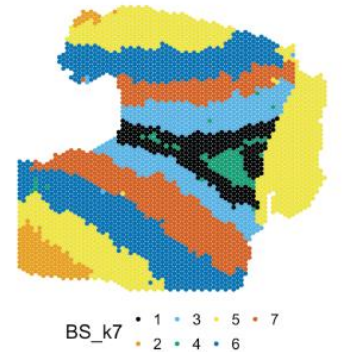
Library size



Mito Ratio



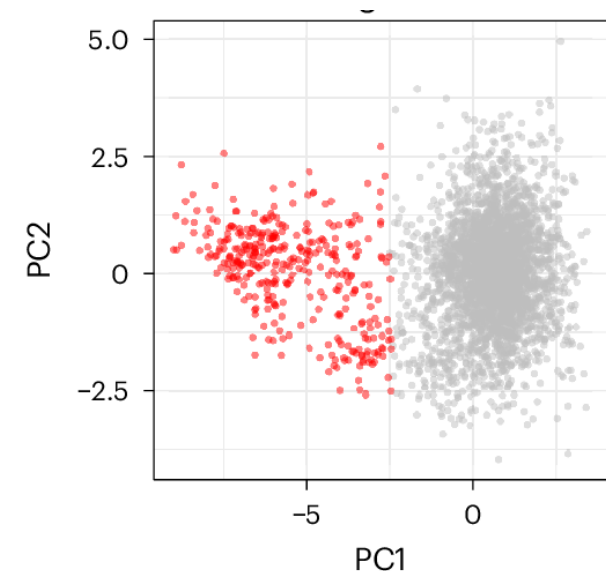
BayesSpace k=7



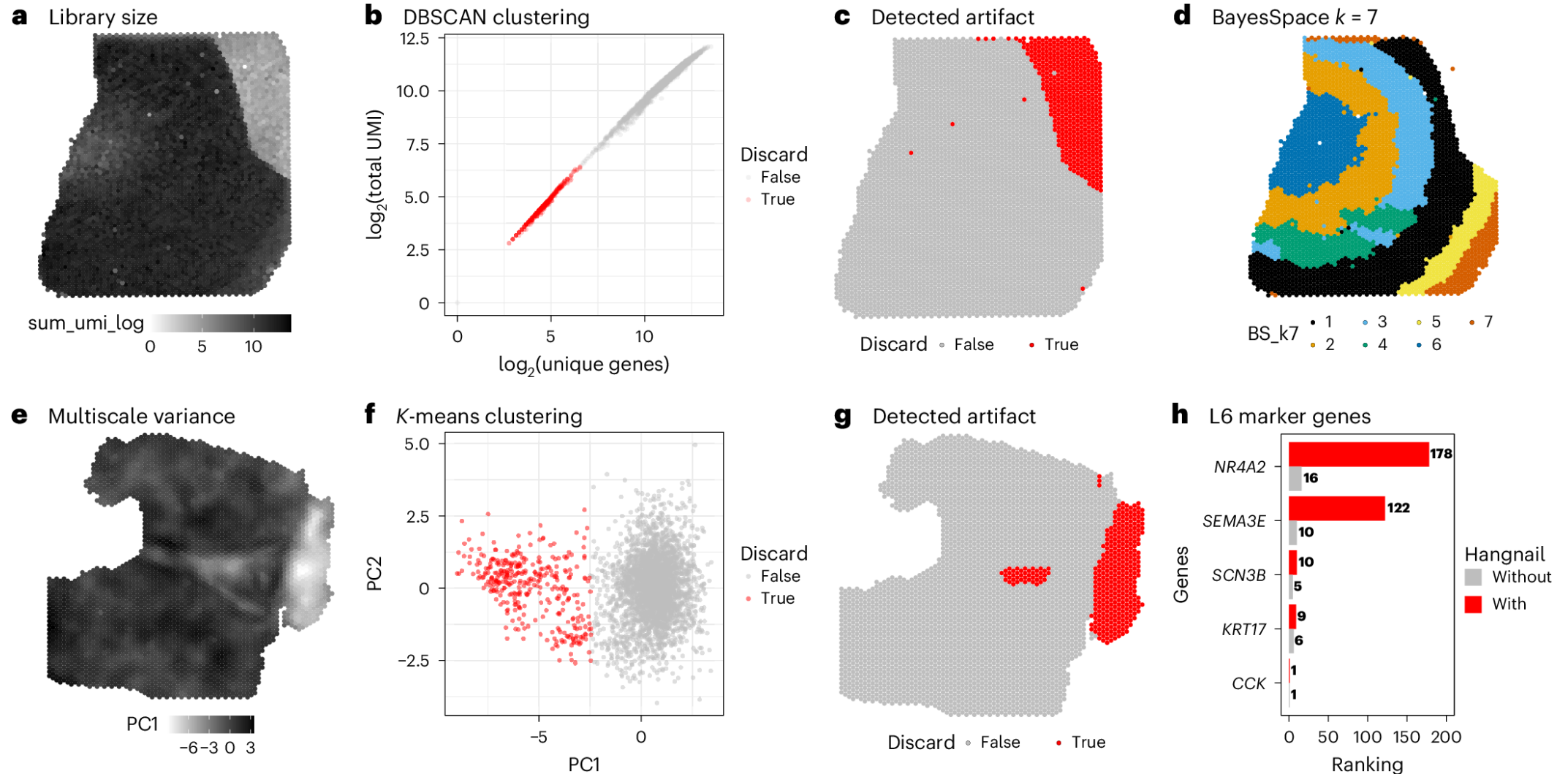
SpotSweeper (region-level artifacts)

1. The k-NN for each spot are identified based on the spatial coordinates
2. For each neighborhood size (i.e., scale), local variance of the mitochondrial ratio is calculated and adjusted for a mean-variance relationship using linear regression
3. Perform PCA on the mean-corrected local variances of all neighborhood sizes
4. Apply k-means clustering (k=2) in the first two PCs to identify regional artifacts compared to high-quality tissue

$$S_i^2(k) = \frac{1}{k-1} \sum_{j \in \text{NN}_k(i)} (x_j - \bar{x})^2,$$



SpotSweeper (region-level artifacts)



Normalization

Sources of variation

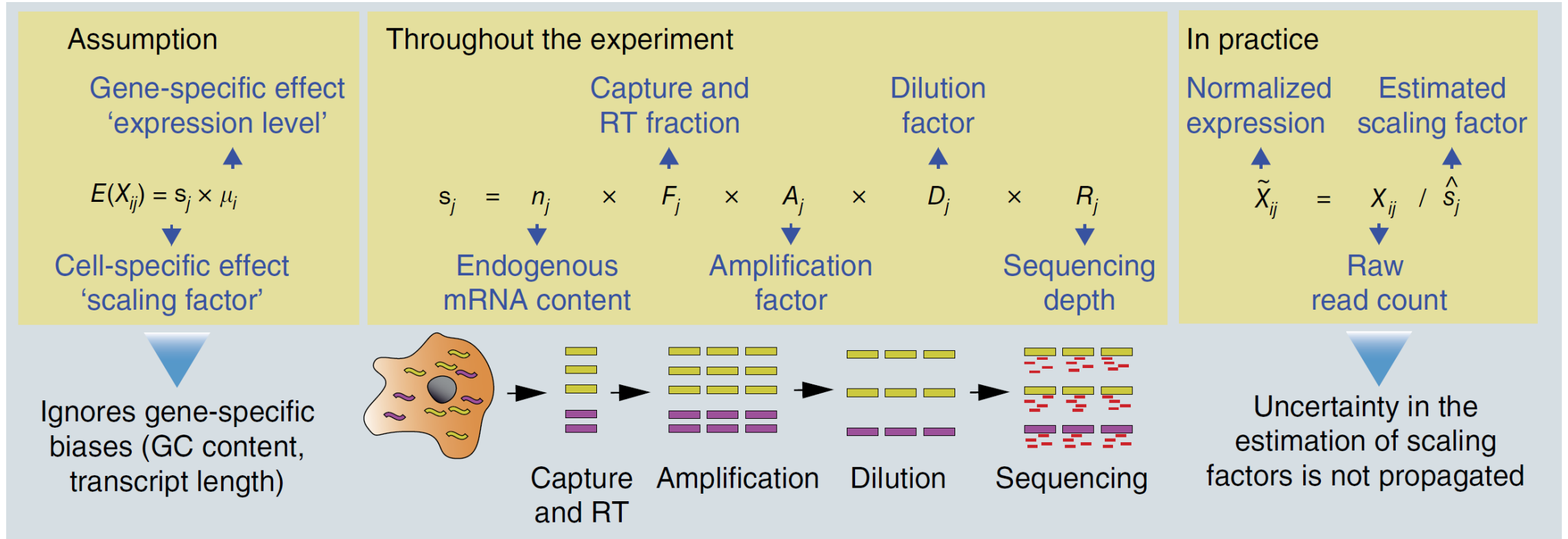
Biological

- Cell type/state
- Cell cycle
- Cell size
- Sex, Age, ...
- ...

Technical

- Cell quality
- Library prep efficiency
- Batch effects
- ...

Normalization (1)

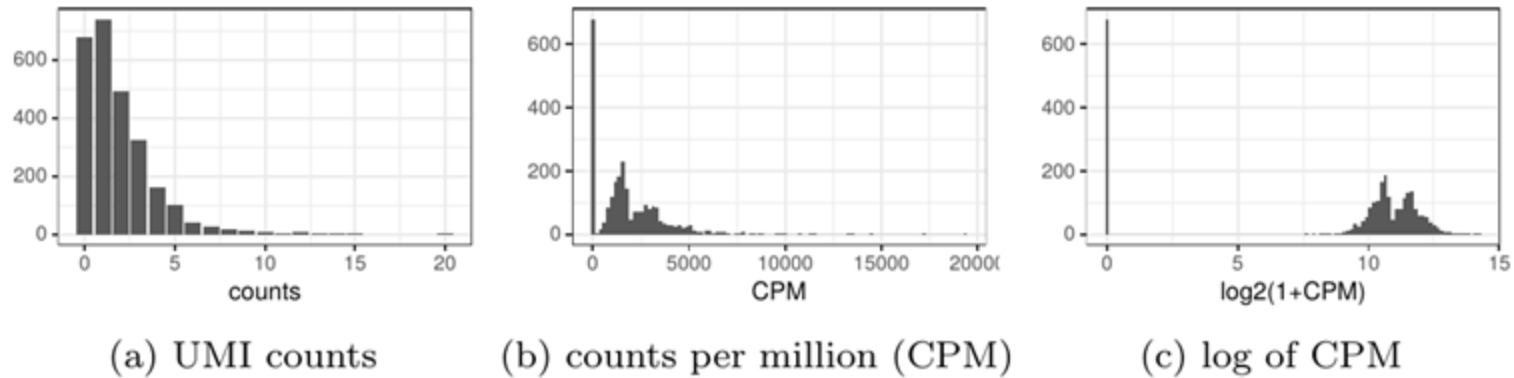


Log-normalization

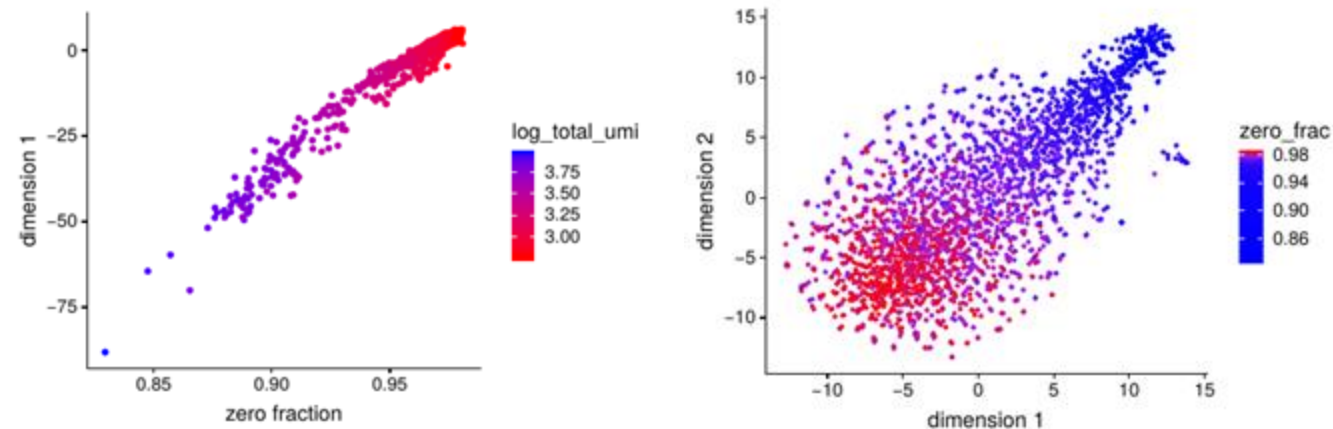
$$Y_{ij} = \log_e \left(\left(\frac{X_{ij}}{\sum_i X_{ij}} \times 10,000 \right) + 1 \right)$$

- Simplest and most commonly-used normalization strategy
- Divide all counts for each cell by a cell-specific scaling factor (i.e. size factor)
- Assumes that any cell-specific bias (e.g., in capture or amplification efficiency) affects all genes equally via scaling of the expected mean count for that cell
- A modified version of CPM normalization

Effect of dropouts on normalization

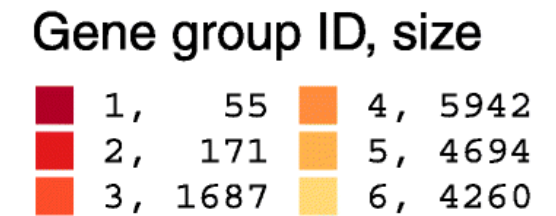
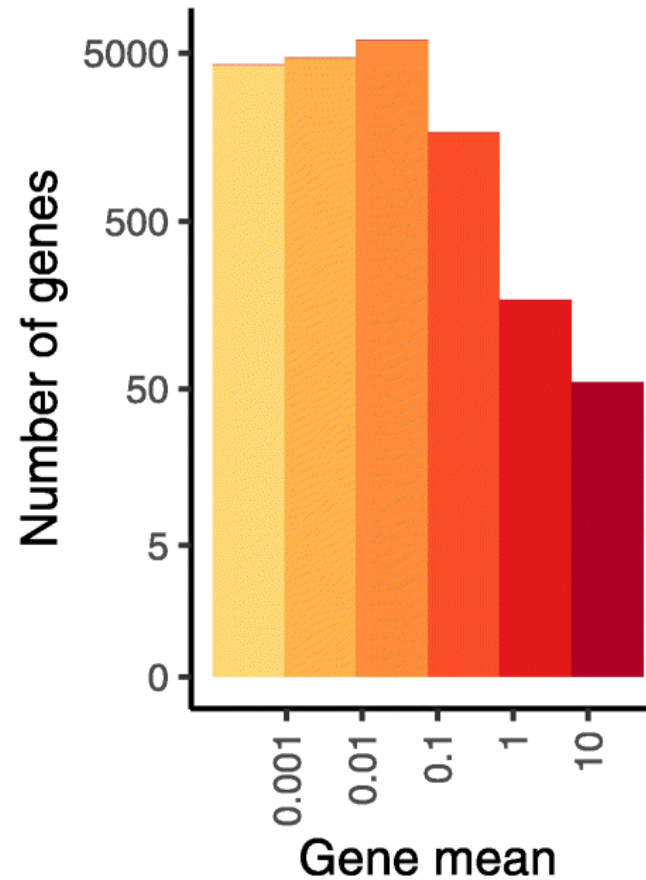
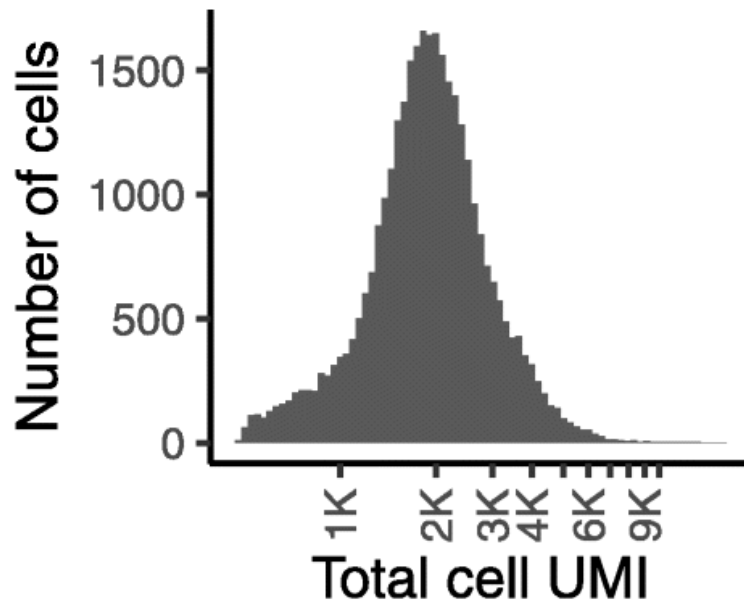


Fraction of zeros become main source of variability



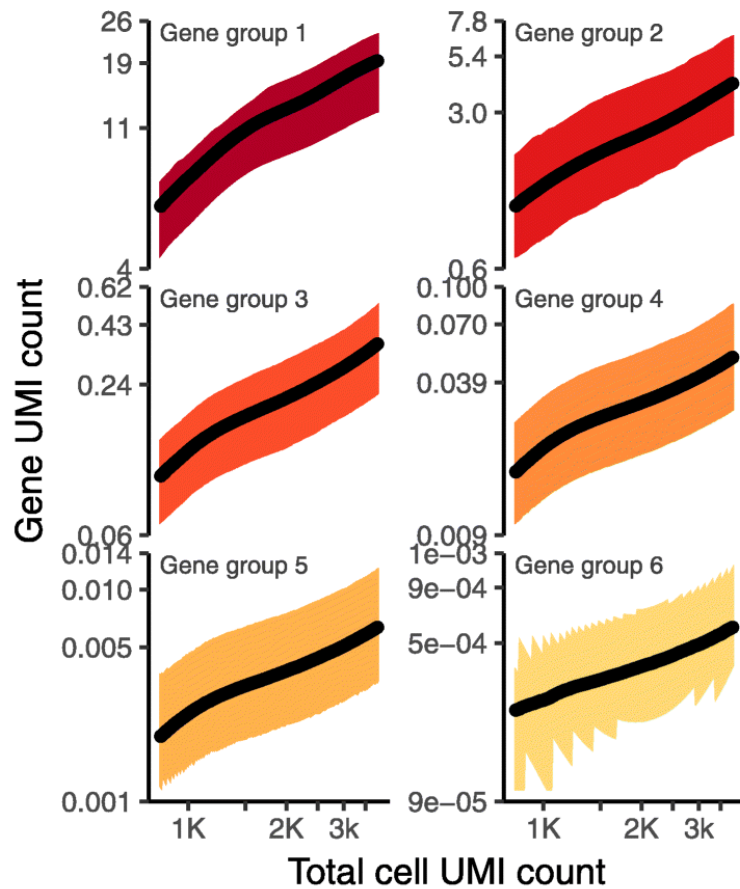
Does log-normalization (scaling) work?

33,148 PBMCs, 10x Genomics
16,809 genes detected ≥ 5 cells

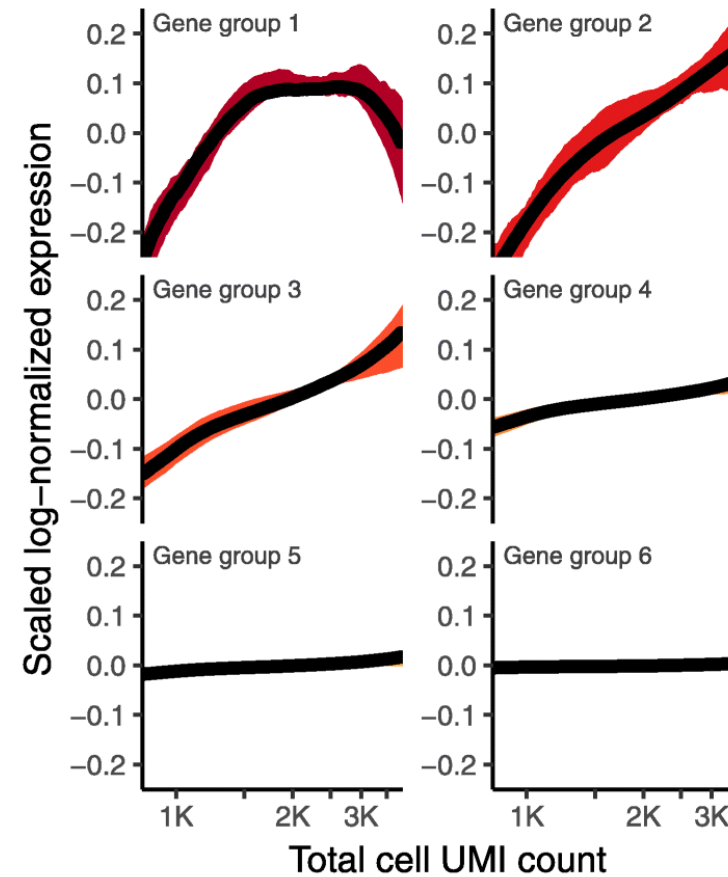


Does log-normalization (scaling) work?

Before normalization



After normalization



Modeling scRNAseq data

- Model the UMI counts for a given gene using a generalized linear model

$$\log(\mathbb{E}(x_i)) = \beta_0 + \beta_1 \log_{10} m + e_i$$

x_i : vector of UMI counts assigned to gene i

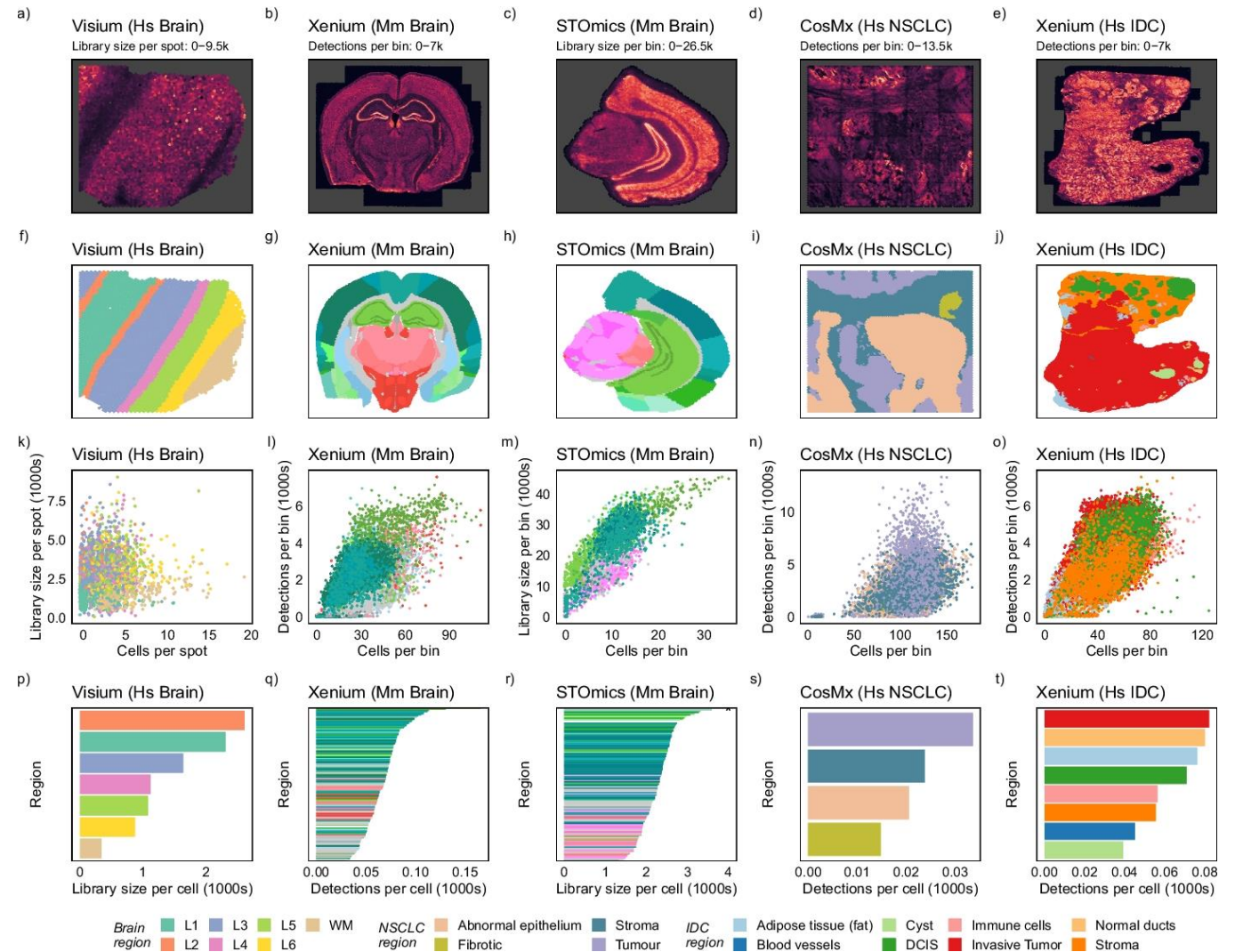
m : vector of molecules assigned to the cells, i.e., $m_j = \sum_i x_{ij}$

e_i : negative binomial (NB) error distribution, parameterized with mean μ and variance $\mu + \frac{\mu^2}{\sigma}$

Allows variation in total library size rather than enforcing it as a constant metric.

Library size $\approx$ biological variation

Library size normalization
assumes that all regions of the
tissue have the same
underlying mRNA abundance



Which normalization method to use?

Bhuva et al. *Genome Biology* (2024) 25:99
<https://doi.org/10.1186/s13059-024-03241-7>

Genome Biology

SHORT REPORT

Open Access

Library size confounds biology in spatial transcriptomics data



Dharmesh D. Bhuva^{1,2,3*}, Chin Wee Tan^{2,3,4}, Agus Salim^{2,5}, Claire Marceaux^{3,6}, Marie A. Pickering⁷, Jinjin Chen^{2,3}, Malvika Kharbanda^{1,2,3}, Xinyi Jin^{2,3}, Ning Liu^{1,2,3}, Kristen Feher^{1,2,3}, Givanna Putri^{2,3}, Wayne D. Tilley⁷, Theresa E. Hickey⁷, Marie-Liesse Asselin-Labat^{3,6}, Belinda Phipson^{2,3†} and Melissa J. Davis^{1,2,3,4,8†}

- Tested the effects of normalization on spatial domain identification

Though SCTransform removes library size effects effectively, their confounding with biology results in removal of biological effects as well.

See also for imaging-based SRT...

Atta et al. *Genome Biology* (2024) 25:153
<https://doi.org/10.1186/s13059-024-03303-w>

Genome Biology

RESEARCH

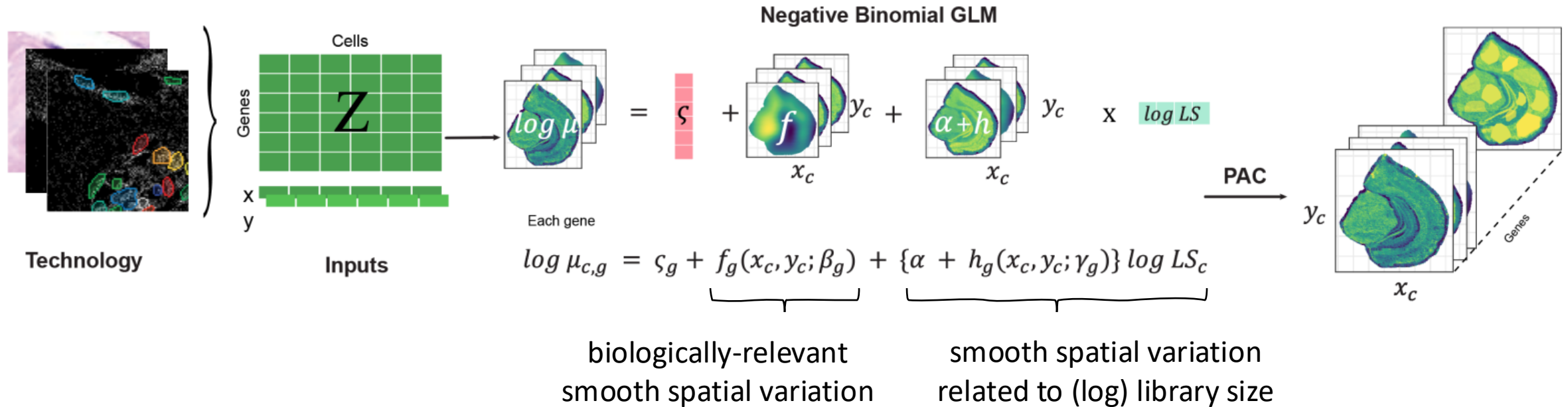
Open Access

Gene count normalization in single-cell imaging-based spatially resolved transcriptomics



Lyla Atta^{1,2}, Kalen Clifton^{1,2}, Manjari Anant^{2,3}, Gohta Aihara^{1,2} and Jean Fan^{1,2*}

SpaNorm

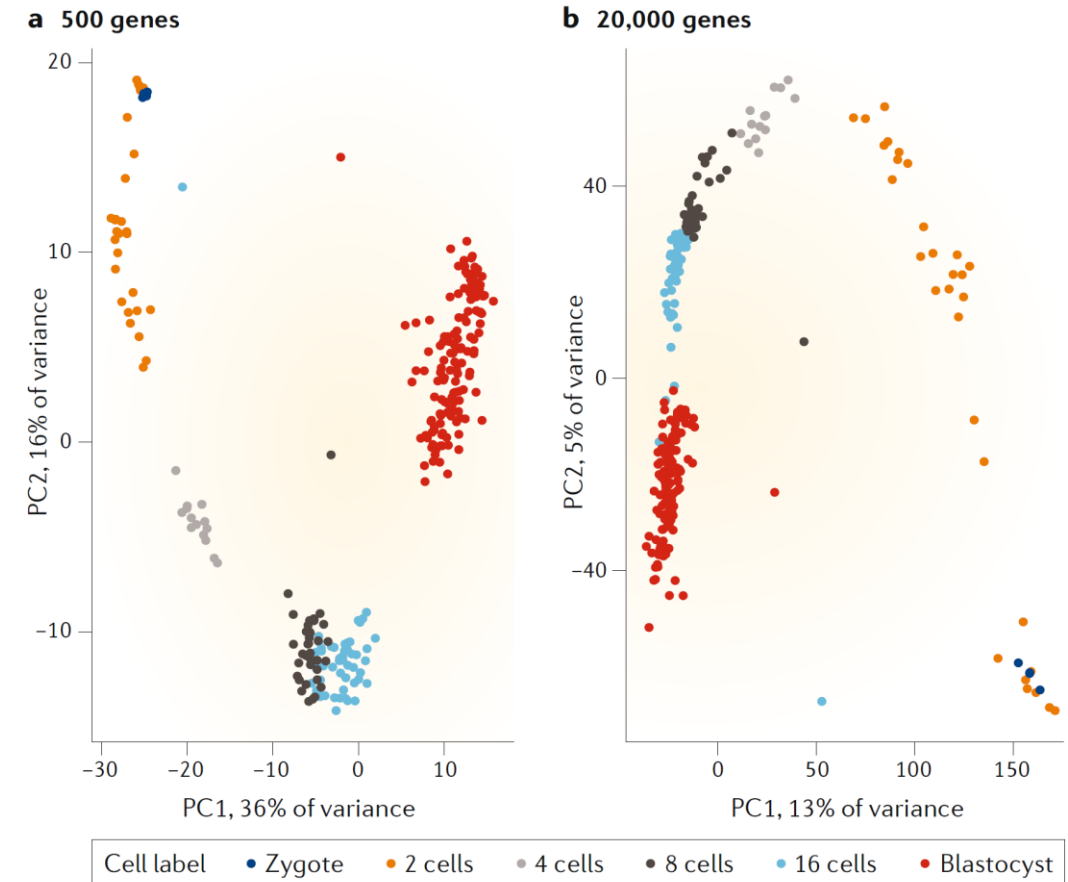


f_g and h_g are two-dimensional, gene-specific spatially-smooth functions constructed using 2D splines

Feature (Gene) selection

Feature selection

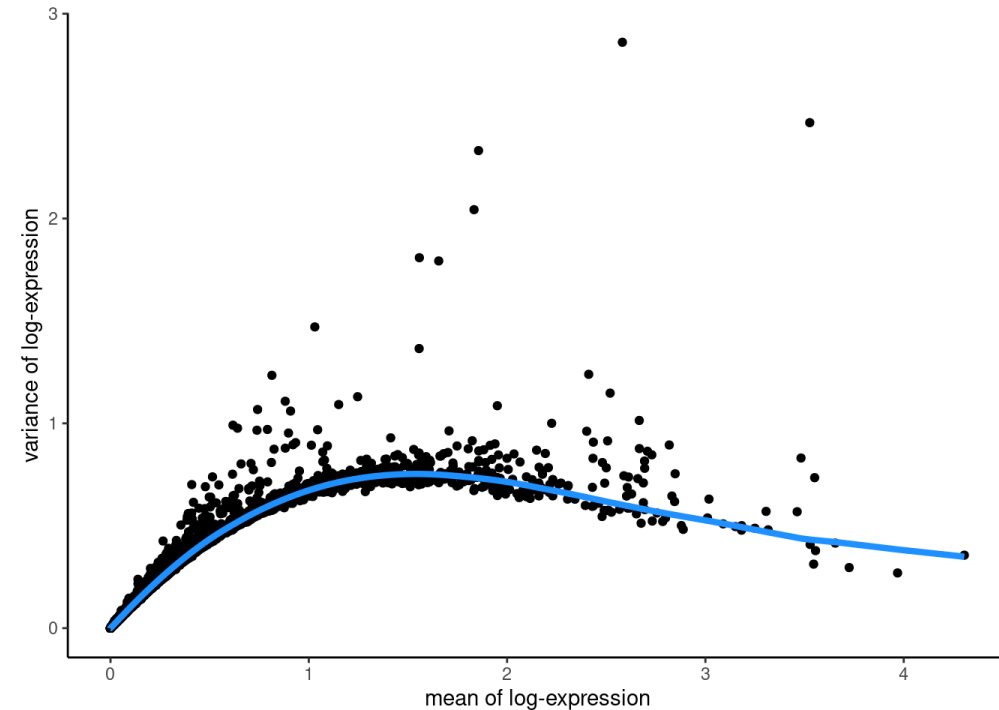
- Curse of dimensionality
 - More features (genes) -> noise dominates distances between samples (cells), effectively all cells get 'same' distance
- Remove genes which only exhibit technical noise
 - Increase the signal:noise ratio
 - Reduce the computational complexity



Feature selection

Highly Variable Genes (HVG)

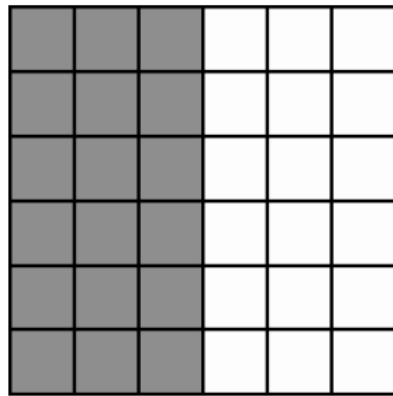
- The simplest approach to quantifying per-gene variation is to compute the variance of the log-normalized expression values across all cells
- We define the **biological component** for each gene as the difference between its total variance and the technical component.
 - Used as the metric for HVG selection (e.g. top 2000 HVGs)



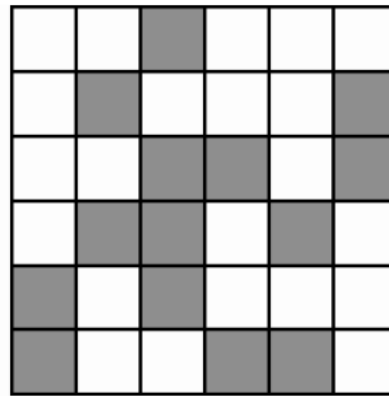
Feature selection

Spatially Variable Genes (SVG)

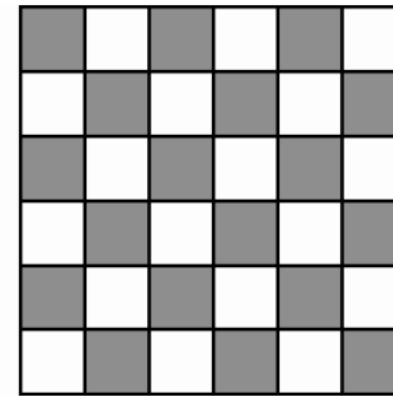
“The first law of geography: Everything is related to everything else, but near things are more related than distant things.” Waldo R. Tobler ([Tobler 1970](#))



Positive spatial
autocorrelation



No spatial
autocorrelation



Negative spatial
autocorrelation

Feature selection

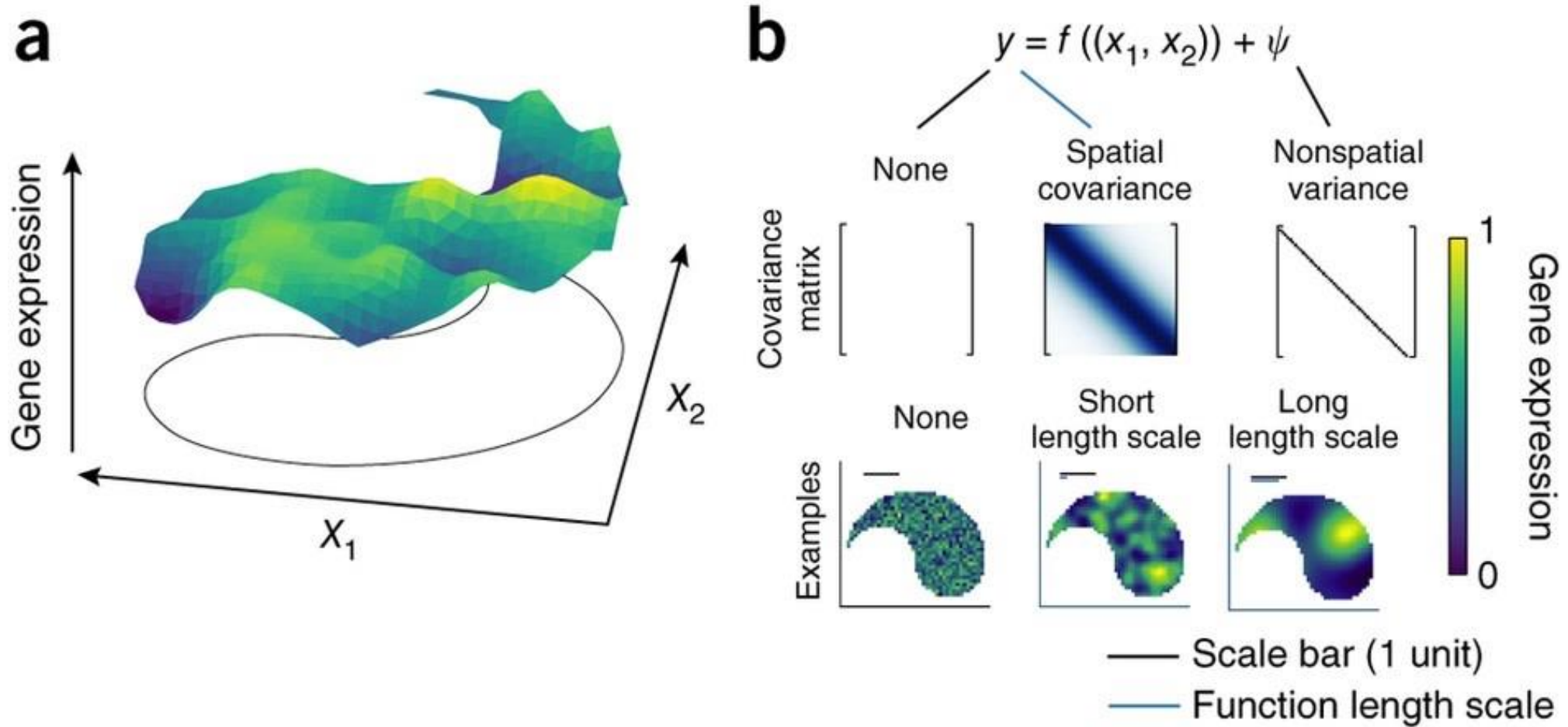
Spatially Variable Genes (SVG)

- HVGs selection methods designed for sc/snRNA-seq data ignores spatial coordinates
- SVGs are genes with a highly spatially correlated pattern of expression, which varies along with the spatial distribution of a tissue structure of interest
- Statistical measures of spatial autocorrelation:
 - Moran's I
 - Geary's C
 - ...

More on spatial statistics for spatial omics, check [Emons et al. NAR 2025](#)

Lecture by Mark Robinson: [slides](#) & [video](#)

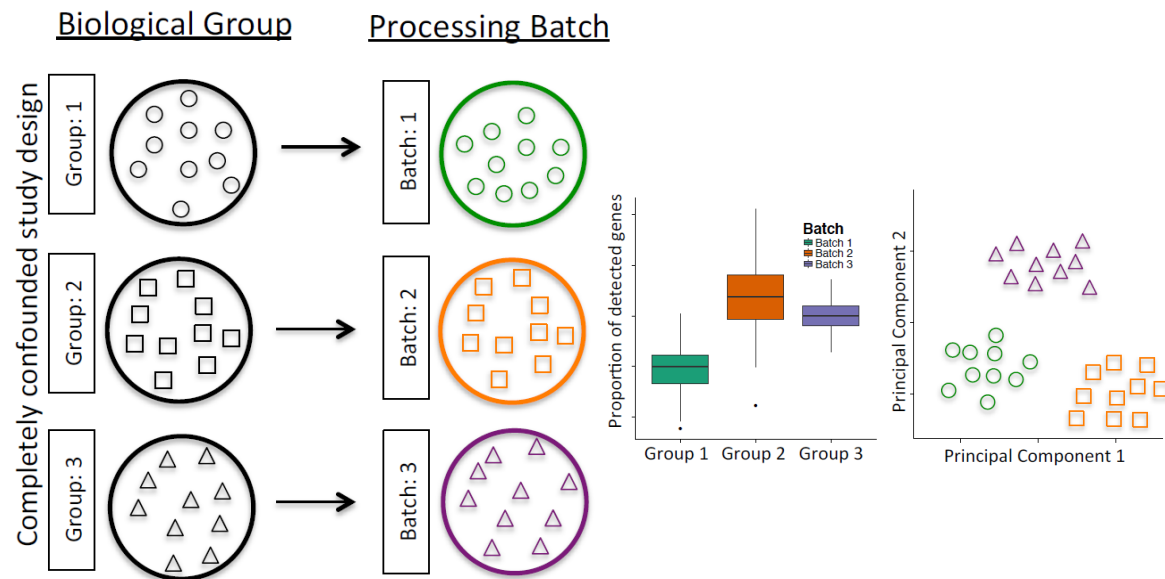
Statistical models of SRT



Batch Correction

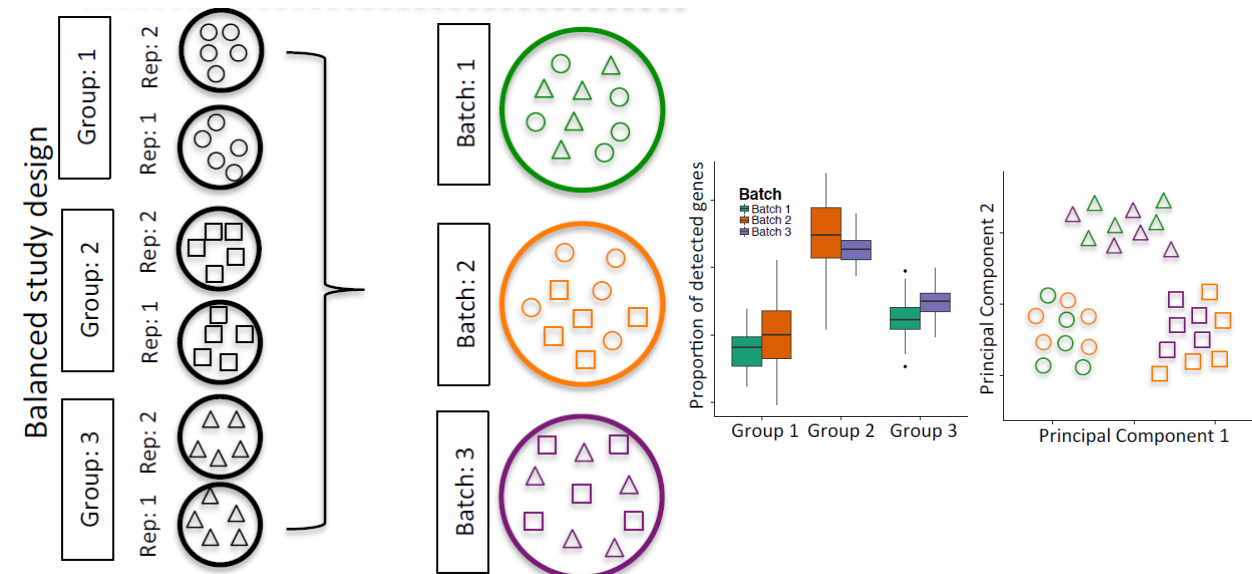
Confounders and batch effects

Confounded design



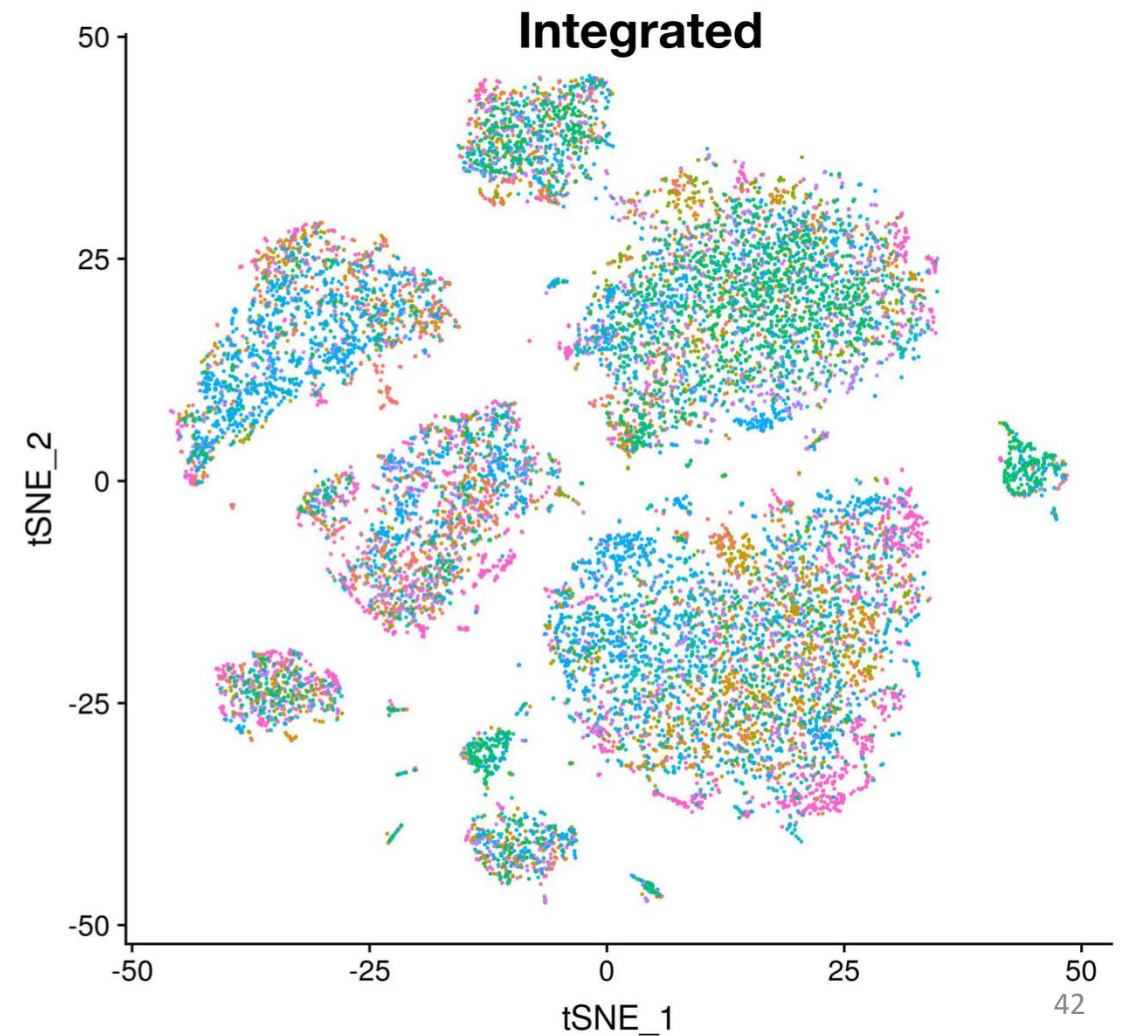
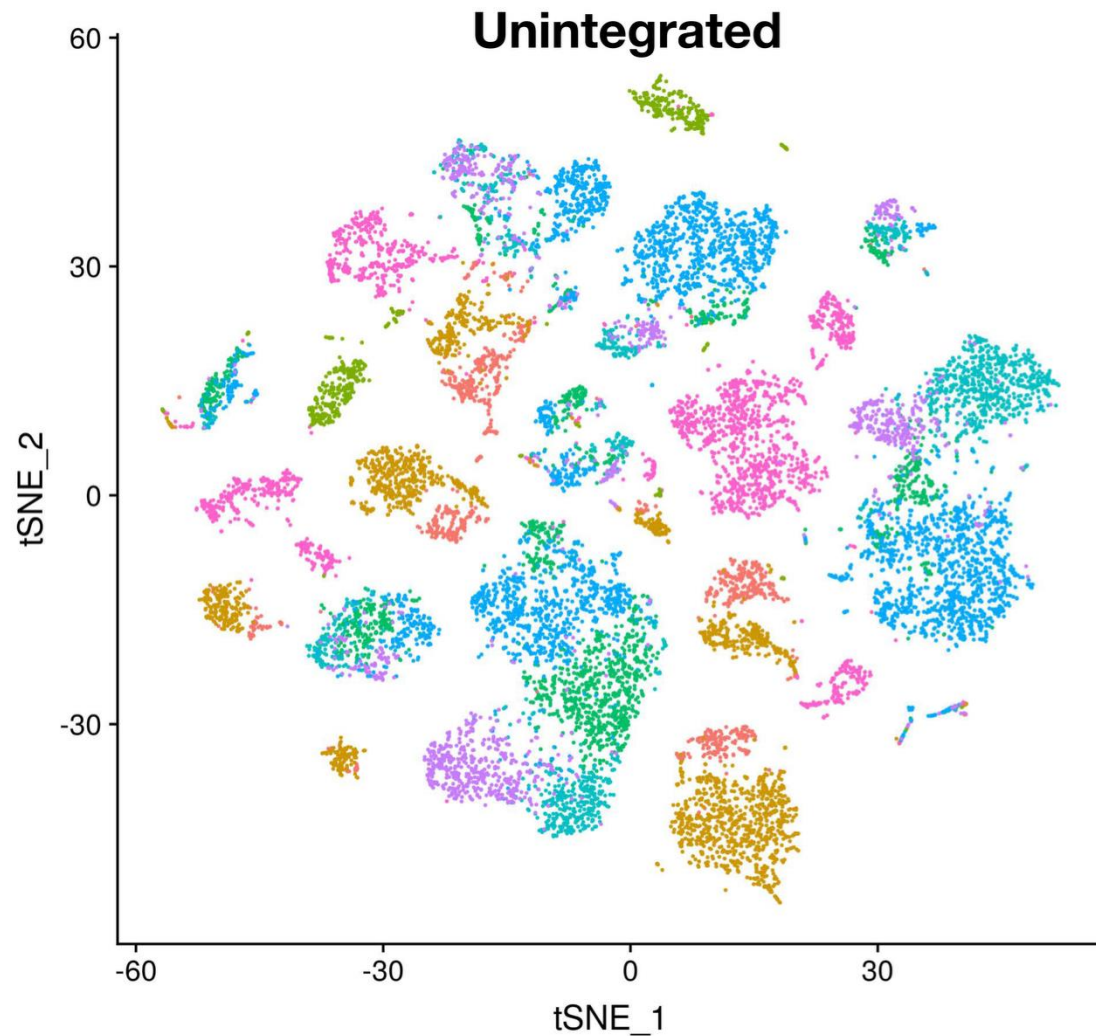
Don't design your experiment like this!!!

Not confounded design



Good experimental design *does not remove batch effects*, it prevents them from biasing your results.

Batch correction



Batch correction methods

- scVI (<https://doi.org/10.1038/s41592-018-0229-2>)
- MNNcorrect (<https://doi.org/10.1038/nbt.4091>)
- CCA + anchors (Seurat v3) (<https://doi.org/10.1101/460147>)
- CCA + dynamic time warping (Seurat v2) (<https://doi.org/10.1038/nbt.4096>)
- LIGER (<https://doi.org/10.1101/459891>)
- Harmony (<https://doi.org/10.1101/461954>)
- Conos (<https://doi.org/10.1101/460246>)
- Scanorama (<https://doi.org/10.1101/371179>)
- scMerge (<https://doi.org/10.1073/pnas.1820006116>)
- ...

Spatially-unaware!

For a recent evaluation of batch effects in spatial transcriptomics:

<https://doi.org/10.1101/2025.03.12.642755>

Summary

- Quality control (QC)
- Normalization
- Feature selection
- Batch correction

Thank You!

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